

Glossary

adapter See expansion card.

American Radio Relay League The ARRL is the national membership association for amateur radio operators. According to the ARRL's official website, <http://www.arrl.org>, the U.S. started licensing amateur radio operators in 1912.

analog Before digital technology, electronic transmission was strictly analog—that is, electrical signals were created, stored, and manipulated in terms of waves and their frequency and amplitudes. TV, telephones, and radios traditionally use analog technology. That's beginning to change now. An analog signal can be represented as a series of sine waves. The term originated because the modulation of the carrier wave is analogous to the fluctuations of the human voice or other sound that is being transmitted. See **sine wave**.

AND gate See logic gate.

assembler A program that converts basic computer instructions into a pattern of 0s and 1s that a computer processor can understand. The result is often called “assembler” or “assembly” language.

assembler language In a computer's assembler language, a language statement generally corresponds to a single instruction. In higher-level languages, like C or Pascal, a language statement can result in multiple instructions. See **assembler**.

atom The smallest particle that can combine with other atoms to form physical elements.

BASIC A simple and popular computer programming language. Originally designed by IBM engineers John Kemeny and Thomas Kurtz in 1963,

BASIC is well known for being easy to learn and widely available for most types of computers.

bit Short for binary digit, this is the smallest unit of data in a computer. It carries a single value, 0 or 1. Eight bits equals a byte. Thirty-two bits is called a “word.” See byte.

board See motherboard.

Boolean algebra See also logic gate. The term “Boolean” is used to describe a type of logic pioneered by early nineteenth-century English mathematician George Boole. It is used to describe a common type of searching on websites where the terms “and” or “or” are used to narrow or expand results. For example, you might search for all websites containing the words Steve AND Wozniak, or choose to search for websites that have either the words Steve OR Wozniak, which would broaden your results considerably. In Boolean algebra, AND and OR are called “operators.”

Electromagnetically, Boolean algebra can be used to describe whether circuit states or memory locations are 1 (On, Charged, or True) or 0 (Off, Not Charged, or False). Engineers can design computers that use an AND gate and an OR gate operation to obtain a result that can be used for the next step in a computational task. To do this, an engineer would have to understand the following basic Boolean algebraic rules:

$$\begin{array}{lll} 0 \text{ AND } 0 = 0 & 1 \text{ AND } 0 = 0 & 1 \text{ AND } 1 = 1 \\ 0 \text{ OR } 0 = 0 & 0 \text{ OR } 1 = 1 & 1 \text{ OR } 1 = 1 \end{array}$$

bus The transmission path of signals in a computer or on a network. Every device connected to the computer along this path, or bus, can potentially receive or generate signals. Devices connected to the computer via expansion slots communicate with the computer via a special expansion bus.

byte A unit of data equal to eight bits. Usually represented in multiples rounded off from the powers of two. For instance, a megabyte (a million bytes) is actually worth 2 to the twentieth power—1,048,576. According to most accounts, IBM engineer Dr. Werner Buchholz invented the term in 1956. See bit.

character A printable on-screen symbol that typically depicts text, a numeral, or a punctuation mark. In computers, there are a limited number of symbols you can use as a character. The reigning standard is called the “ASCII set” (pronounced ASK-KEE), from the American Standards Committee.

chip Short for microchip, this term refers to the amazingly complex and tiny modules that contain logic circuitry that perform functions or act as memory for a computer. Most typically, a chip is manufactured from a silicon wafer and then etched with circuits and other devices in a clean, controlled environment. Chips are sometimes also referred to as “ICs,” or integrated circuits.

chip set A group of integrated circuits (microchips) that can be combined to perform a single function. They are usually sold in a unit—that is, a set.

compiler A special program that takes statements written in a certain computer language and translates them into machine language that a computer processor can understand.

CPU Short for central processing unit, this is the chip or, previously, the set of chips that contained all the logic circuitry in a computer responsible for running computer program instructions. These days, it is more common to call a CPU a “processor” or a “microprocessor.”

current Measured in amperes (amps, or A), current is the flow of electrons or other electrical charge carriers. It can either be direct (DC), flowing in the same direction at all points in time, or alternating (AC). The number of AC signals, in which the flow of electrons changes frequency periodically, is measured in cycles per second (hertz) and is called its frequency. See **frequency**.

debug The process of locating and fixing (or bypassing) bugs and other errors in computer program code or on a computer hardware device. The term “debugging” refers to a procedure that begins with naming a problem, isolating its source, and fixing the problem at its source. This is a necessary procedure in the building and design of any computer software or hardware program.

digital A term describing electronics that creates, stores, and manipulates data as defined in only one of two possible states—1 or 0 (On or Off). Each one of these state digits is called a “bit,” and a string of eight bits together is called a “byte.” See **bit**; **byte**.

diode An electronic device that restricts current flow to one direction only.

diskette A removable storage medium for personal computers. Until the early 1990s, most personal computers used a “floppy” format, 5.25-inch flexible magnetic disks set inside plastic envelopes. Most personal computers these days use a smaller, rigid 3.5-inch diskette format.

DRAM Dynamic random-access memory (DRAM) is today the most common kind of memory chip available for computers. The term “random access” refers to a CPU’s ability to rapidly find data (in the form of 1s and 0s) stored on computer memory chips—and do so directly. Before DRAM, CPUs had to access the data in memory sequentially, by starting at the beginning of the stored data and proceeding forward. The term “dynamic” means the memory chip has to be continually refreshed electrically to retain its information.

dynamic random-access memory See DRAM.

EEPROM Short for electrically erasable programmable read-only memory, this is a type of read-only memory that can be erased and reprogrammed with new data for a limited number of uses, typically by applying high electrical voltage to the chip. See EPROM; PROM.

electron A negatively charged subatomic particle. In electrical conductors, electronic current is the result of moving electrons from atom to atom—that is, from negative to positive poles. In semiconductor materials, current also results from moving electrons.

electronic gate See logic gate.

ENIAC The Electronic Numerical Integrator And Computer was one of the first true computers in the world. The U.S. Army completed it in 1946—its purpose was to calculate ballistic firing tables for the Ballistics Research Laboratory. The ENIAC (pronounced EENIE-ACK) was built at the University of Pennsylvania by researchers J. Presper Eckert and John William Mauchly.

EPROM Short for erasable programmable read-only memory, this is a type of programmable read-only memory chip with contents that can be erased and reused. Old data is erased if it is exposed to an especially intense ultraviolet light. See EEPROM; PROM.

expansion card Also simply called a “card,” “board,” or “adapter,” these circuit boards typically bring increased functionality to a computer. A user plugs an expansion card into one of the computer’s expansion slots, effectively adding circuitry to the computer. See motherboard; slot.

expansion slot Also called a “slot,” this is a connector that allows the addition of expansion boards (or cards), circuit boards that bring more capability to a computer. For instance, a user might plug in an expansion card to add a high-end data plotter or scanner to a computer. These days, all desktop computers come with expansion slots that allow you to increase functionality in this way.

floppy See diskette.

FORTRAN Short for FORMula TRANslation, this computer language was designed especially for use by mathematicians, engineers, and scientists. These days, such scientific users typically instead use the C programming language.

frequency Measured in hertz (Hz), frequency is the number of complete cycles per second. For instance, a current measured at 60 cycles per second would be rendered 60 Hz. Megahertz (MHz) and gigahertz (GHz) represent millions and billions of cycles per second, respectively.

gigabyte A gigabyte is an amount of computer data roughly equal to a billion bytes—that is, 2 to the thirtieth power, or 1,073,741,824. See bit; byte; kilobyte.

hard disk drive Also referred to as “hard drive,” “disk drive,” and “hard disk,” this refers to a permanent storage device used by computers. These days, computers can store billions of bytes (gigabytes) of data on their built-in hard disks. If you could see one inside, a hard disk is actually like a mini stack of disks, not unlike the music albums that preceded CDs. They have concentric tracks on their disks, with stored data located throughout the track. The drive typically has two heads, one on each side of each disk, reading data from or writing data to a disk. If you are saving a letter you wrote on a word processor, you are saving it to a hard disk.

hertz A unit of frequency equal to a cycle per second. Named for the German physicist Heinrich Hertz.

hexadecimal A Base 16 system commonly used by today’s digital computers, which work with binary digits (1 and 0) and bytes (eight 1s and 0s, or bits) of information at a time. Two hexadecimal digits can represent a byte, as follows:

Binary	Decimal	Hexadecimal	Binary	Decimal	Hexadecimal
0	0	0	1001	9	9
1	1	1	1010	10	A
10	2	2	1011	11	B
11	3	3	1100	12	C
100	4	4	1101	13	D
101	5	5	1110	14	E
110	6	6	1111	15	F
111	7	7	10000	16	10
1000	8	8	10001	17	11

infinite loop Often also called an “endless loop,” this is a piece of coding that accidentally or by design repeats indefinitely.

instruction This is a key term in computer technology. It is an order created by a computer program delivered to a computer processor. Each instruction, at its most basic level, is just an order for the computer to do something (like “add” or “subtract”) with the 1s and 0s that make up computer data. See **assembler language**; **register**.

interrupt This refers to a signal that comes from a device attached to a computer—or from a program running on that computer—that causes the CPU system software to stop and pay attention to what should be done next. Almost all computers today use interrupts. That is, they run whatever program they’re running instruction by instruction until they are “interrupted” by a device or another program. For instance, if you hit the G key while a program is running, the system will pause, notice that the keyboard has interrupted it, and run the program that will display “G” on-screen.

kilobyte A unit of computer storage equal to approximately 1,000 bytes of data—more precisely, 2 to the tenth power, or 1,024 bytes.

logic gate A single computer circuit that has several points of input but only one point of output. It is an elementary building block of a circuit. Most logic gates have two inputs and one output.

At any given moment, every terminal is in one of the two conditions—low (0) or high (1)—defined by the voltage level. The state, 0 or 1, changes often as data is processed. For example, the AND gate is called that because if 0 is false and 1 is true, the gate acts the same way as the standard AND operator in Boolean algebra.

With an OR gate, the output is true (or 1) if either or both of the inputs are true (or 1). If both inputs are false (0), then the output is false (0).

The XOR (exclusive-OR) gate acts in the same way as the logical “either/or.” The output is “true” if either, but not both, of the inputs is “true.” The output is “false” if both inputs are “false” or if both inputs are “true.”

A logical inverter, sometimes called a NOT gate to differentiate it from other types of electronic inverter devices, has only one input. It reverses the logic state.

The NAND gate operates as an AND gate followed by a NOT gate. It acts in the manner of the logical operation “and” followed by negation. The output is “false” if both inputs are “true.” Otherwise, the output is “true.”

The NOR gate is a combination OR gate followed by an inverter. Its output is “true” if both inputs are “false.” Otherwise, the output is “false.”

The XNOR (exclusive-NOR) gate is a combination XOR gate followed by an inverter. Its output is “true” if the inputs are the same, and “false” if the inputs are different.

Using combinations of logic gates, complex operations can be performed. In theory, there is no limit to the number of gates that can be arrayed in a single device. But in practice, there is a limit to the number of gates that can be packed into a given physical space. Arrays of logic gates are found in digital integrated circuits (ICs). As IC technology improves, the physical space it takes up becomes smaller and smaller. That means faster chips in smaller packages—and increasing computer power at decreasing prices. See **Moore’s Law**.

machine code The basic-level language that a computer can understand, this refers to a stream of binary digits—0s and 1s—or bits. See **bit**; **byte**.

memory The electronic holding place for instructions and data that your computer needs to reach quickly. Typically referred to as “RAM” (short for random-access memory), memory is typically located on a set of microchips located physically close to the computer processor. When you turn a computer off, all information held in RAM disappears.

Moore’s Law Intel founder Gordon Moore in 1964 made the following now-famous observation: that due to improvements in manufacturing, every eighteen months engineers would be able to double the number of transistors on a chip. Moore’s Law has held true to this day.

motherboard The physical layout inside a computer that contains its basic circuitry and components. The motherboard most typically contains the CPU, the main system memory, the basic input/output system (BIOS), a group of expansion slots, and additional interconnection circuitry. Also sometimes referred to as the “main board” and the “system board.”

NOR gate See **logic gate**.

OR gate See **logic gate**.

oscilloscope A laboratory instrument commonly used to display and analyze the waveform of electronic signals. On a screen, the device draws a graph of voltage over time.

processor The logic circuitry in a computer that responds to a computer instruction. Generally people use the term “processor” to refer to a computer’s central processing unit (CPU). Another common term for the CPU is “microprocessor.” See **CPU**.

PROM Short for programmable read-only memory, this is a type of computer chip with data that can only be changed with a special machine. Such a machine, often called a “PROM programmer,” actually blows a fuse on the chip—hence the term “burning a PROM.” See EEPROM; EPROM.

RAM Short for random-access memory, this is the type of memory chip a computer uses for short-term storage and calculation. This kind of “memory” is not to be confused with the permanent storage a hard disk or CD-ROM drive provides. RAM chips lose their contents whenever you power down the computer. See *memory*.

register In a computer processor, a register is a holding place for any kind of data, including a storage address, individual characters, or a computer instruction. For example, a computer instruction might command that the contents of two registers be added together. A register is typically large enough to hold a 32-bit instruction, though there are smaller registers, such as half-registers, in some computer designs.

resistance Denoted by the representation R , this is the opposition a given substance offers to the flow of current. Measured in ohms.

resistor An electronic component that controls the flow of current in a circuit by resisting, or turning away, electricity. Typically, resistors are mounted on a printed circuit board or built into a chip.

ROM Built into every computer, ROM, short for read-only memory, is computer memory that contains data that can only be read. It is designed to store data permanently, and not be erased or modified by a user. A ROM chip contains the program that allows a computer to be restarted and still remember its basic settings every time. Unlike the computer’s RAM (random-access memory), the data on this chip stays intact even when you turn the power to the machine off. The ROM is typically powered by a small long-life battery. See EEPROM; EPROM; PROM; RAM.

signal Most simply, this is an electric current or field used to carry data from one place to another. A direct current (DC) signal that can be switched on and off is a simple form of carrying information—it’s how the early telegraph signals worked. A more complicated signal consists of an alternating current (AC) to carry more than one stream of data at a time.

sine wave The most familiar alternating current waveform, varying with time. A waveform is a pictorial representation of how alternating current (AC) varies over time.

slot See expansion slot.

storage Refers to the place in a computer where data is held in electromagnetic or optical form for access by a computer processor. The term “primary storage” generally refers to the place in memory where data is held; the term “secondary storage” generally refers to permanent data holding on hard disks, tapes, and other storage media.

transistor A tiny device for regulating electronic signals. Invented by three scientists at Bell Laboratories in 1947, this was a key invention that enabled computers and computerized devices. Before transistors, vacuum tubes were in use—but they quickly became obsolete after the transistor was widely available for the purpose of regulating current (voltage). Transistors act as incredibly tiny and effective switches for electronic signals. See vacuum tube.

transistor circuit See transistor.

vacuum tube Also known as an “electron tube,” a vacuum tube was once commonly used to amplify electronic signals. It is now mostly obsolete, having been replaced in electronics by the transistor. See transistor.

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I have to thank those responsible for my successes in life. To Miss Skrak for seeing so much in me. To Mr. McCollum for finding so much valuable education beyond the school he worked for. To Steve Jobs for wanting to do Great and Big things. To Randy Wiggington, Chris Espinoza, Dan Sokol, Bill Fernandez, and the Homebrew Computer Club members for the whole appreciation of a

computer for people. To all my HP friends, including Stan Mintz and Peter Dickinson, for a great environment in which an engineer could develop. Most of all, thanks to Allen Baum, who was involved in some way in so many of the big steps in my computer life. And for his parents who appreciated jokes and humor and had such good values in life. My memory of them still brings tears.

I must thank my first wife, Alice, without whom Apple would never have happened for me; Candi, my second wife, for the most wonderful creations of Jesse, Sara, and Gary; and my third wife, Suzanne, for Hard Rock Cafes and bungee jumping and for being so wonderful and decent.

Friends who made this possible for me, digging up needed photos and reminding me of stories, included Laura and Dan and Alex. Sharon was the most loyal in all cases, always looking out for me and making sure needed things got done.

• • •

Gina thanks her friend Michele Earl for introducing her to Steve at a rock concert. Within a week of our meeting, the *iWoz* book proposal was in to our phenomenal agent, John Brockman.

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also to Gina's half-sister, Isabella, her husband, Roger, and Gina's much-loved nieces, Victoria and Alexandra. Thanks also to Maria Lopez and Gina's in-laws Lisa and Henry Schaefer for their babysitting support during this enormous project. And finally, Gina is forever grateful to those who have stood beside her the longest: Sister Laura Saucedo and Brother Keith Prewitt. Love and peace to you all.

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Steve and Gina both thank the two restaurants they dominated weekly while doing fifty-six two-hour interviews. The first half of the book was completed at Pearl's, in the West Portal District of San Francisco. The second was completed at The Hick'ry Pit in Campbell, California, where we especially thank our waitress Racquel and her boss, Brian, for bending the rules for us every now and then.

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Thank you, readers. We hope you enjoy Steve's adventure as much as he enjoyed telling it and Gina enjoyed hearing it!



People used to go to Caltech games just to see my dad play. Here he is in uniform. *(Photograph courtesy of Margaret Wozniak)*



Here's my mom and dad's wedding picture. *(Photograph courtesy of Margaret Wozniak)*



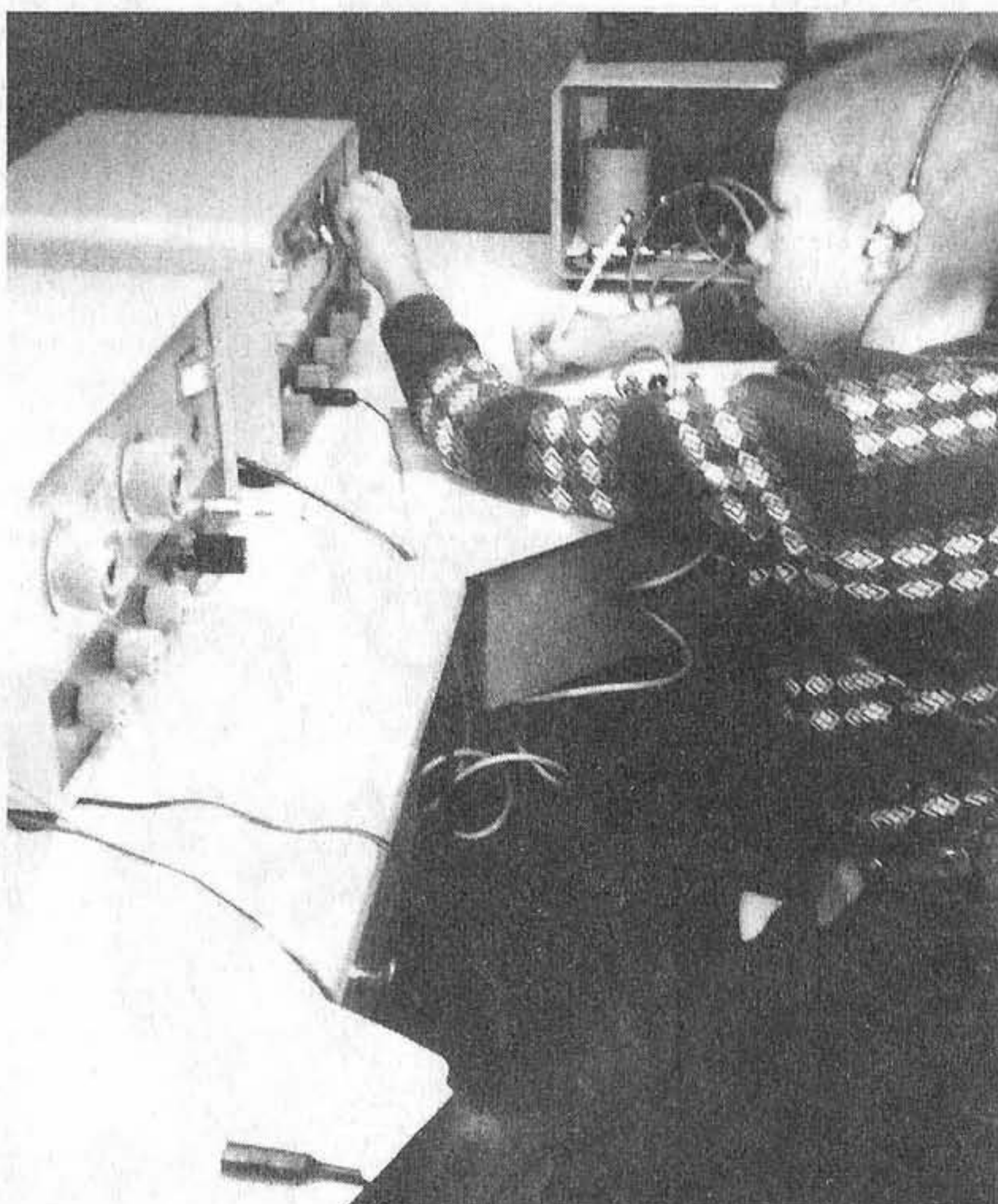
They tell me I was reading at three.
(Photograph courtesy of Margaret Wozniak)



My dad and the three of us siblings. From left: me, Mark, and Leslie. *(Photograph courtesy of Margaret Wozniak)*



Here I am at eleven, in Little League. (*Photograph courtesy of Margaret Wozniak*)



At eleven, I was one of the youngest ham radio operators in the world. But I got bored. No one my age to talk to! (*Photograph courtesy of Margaret Wozniak*)



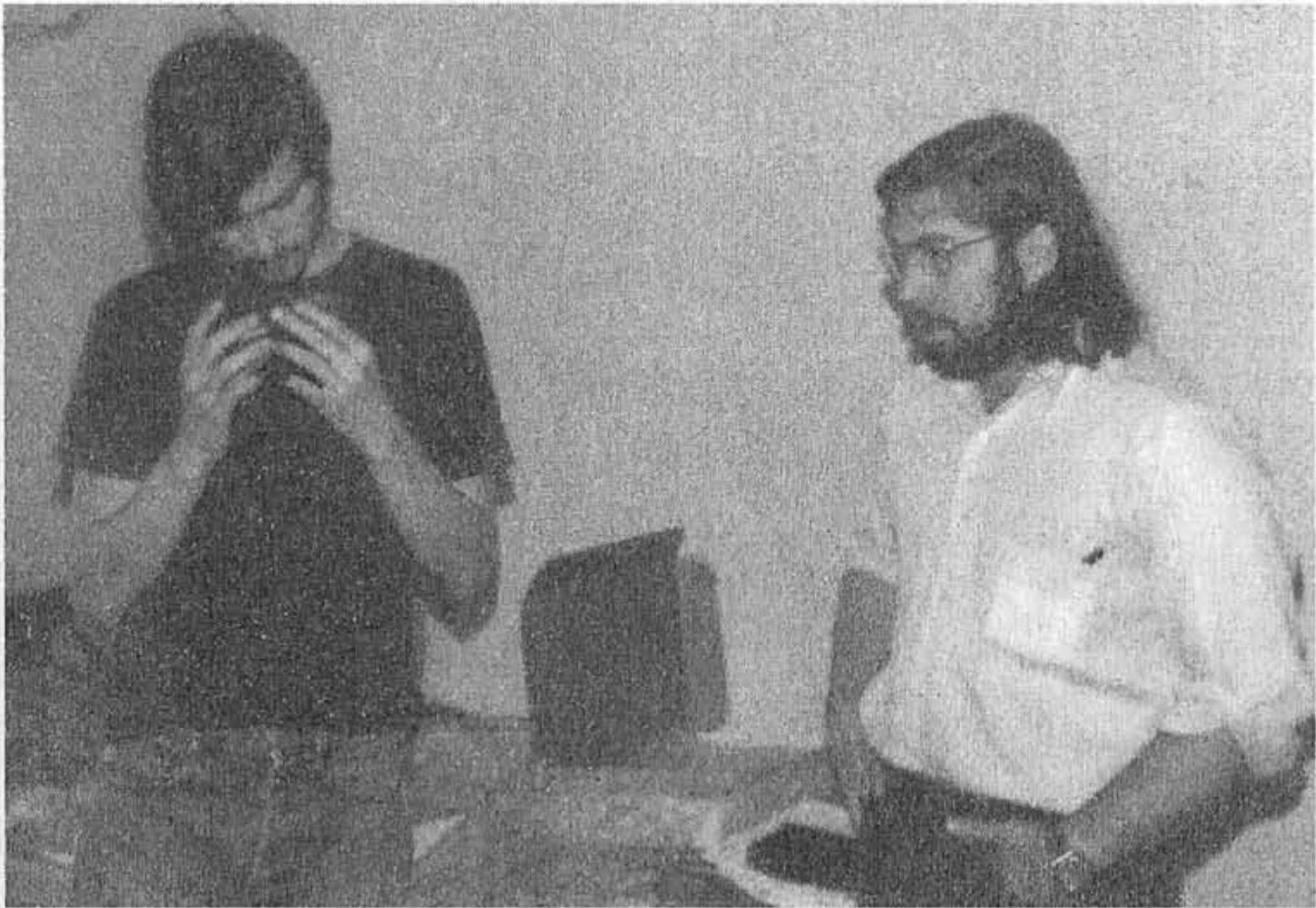
Here I am at thirteen showing off my science-fair-winning Adder/Subtractor.
(*Photograph courtesy of Margaret Wozniak*)



Here I am at thirteen in 1963, graduating from junior high. (*Photograph courtesy of Margaret Wozniak*)



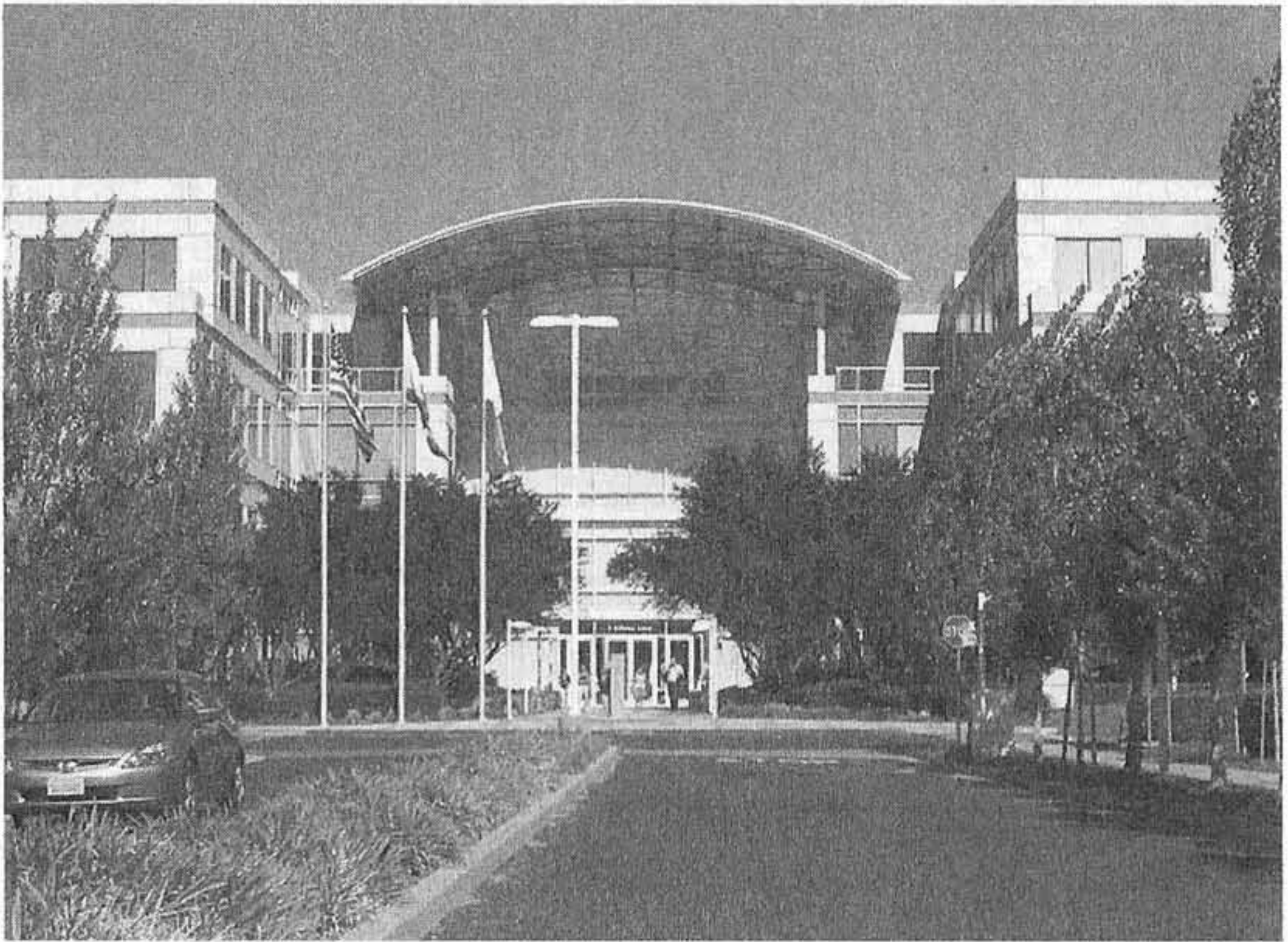
Here's Allen Baum and me (left) showing off our "Brazilian Best Wishes" banner, to shake things up at our old high school—then Steve Jobs'. Allen and I had graduated four years earlier. *(Photograph courtesy of Margaret Wozniak)*



Young Steve Jobs and me in 1974, with the "Blue Box" I designed. *(Photograph courtesy of Margaret Wozniak)*



In 1978, Apple got a real office! Here I am testing some new hardware. *(Photograph courtesy of Dan Sokol)*



Oh, how times have changed! Apple's headquarters at 1 Infinite Loop in Cupertino are pictured here. *(Photograph courtesy of Wikipedia)*



My friend Dan Sokol gave me this framed Apple I circuit board as a thirtieth birthday gift. It was displayed in the Apple lobby for years. *(Photograph courtesy of Dan Sokol)*



Before we had the volume to pay for plastic cases, a lot of our customers would cover the Apple I board with a wooden case, often made of Koa wood. *(Photograph courtesy of Wikipedia)*



The Apple II—my creation—is the computer that changed the world. So they tell me. *(Photograph courtesy of Wikipedia)*



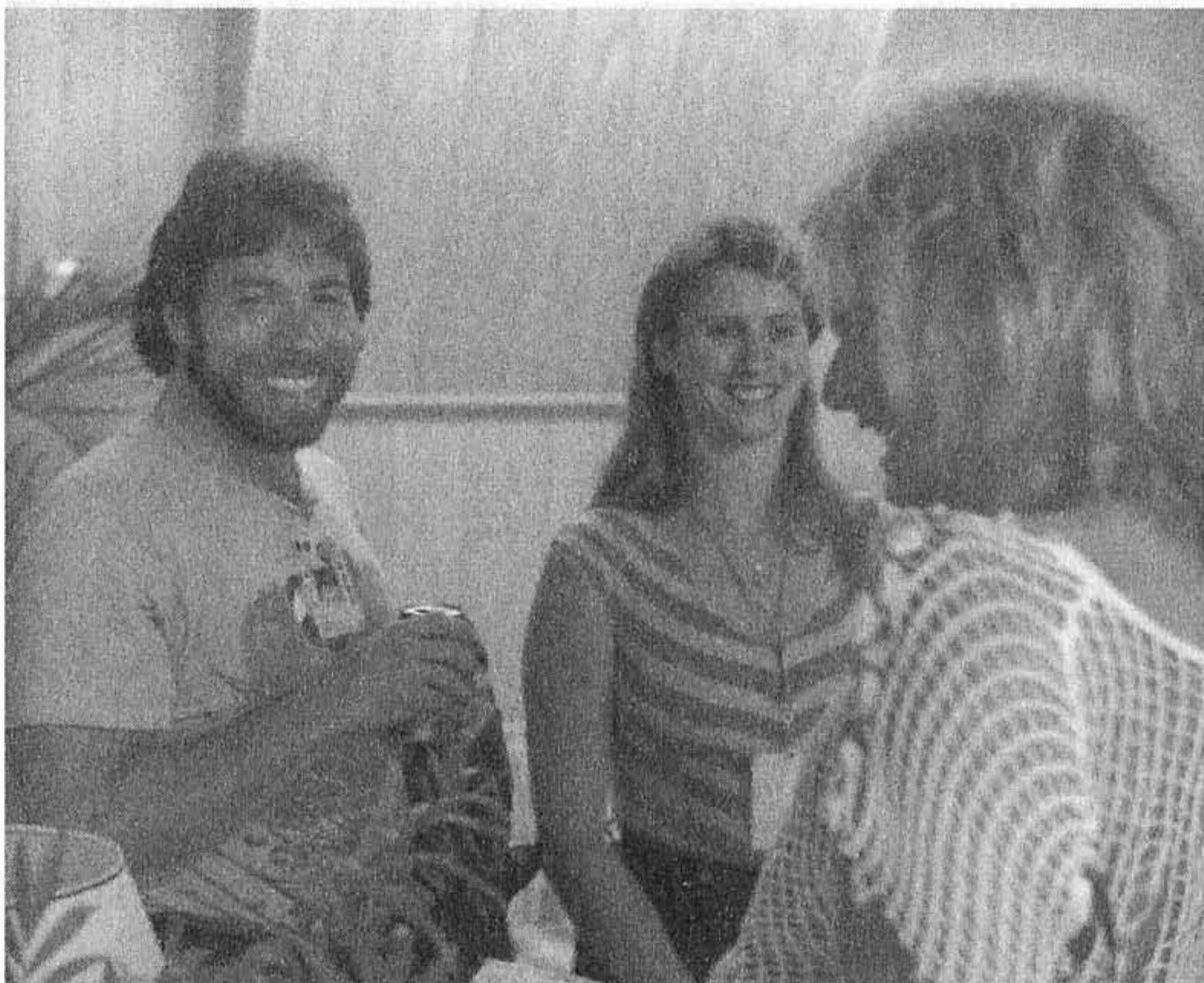
The Apple III was the computer that was designed by committee. Apple pushed it like crazy, but the majority of people still wanted the Apple II. (*Photograph courtesy of Wikipedia*)



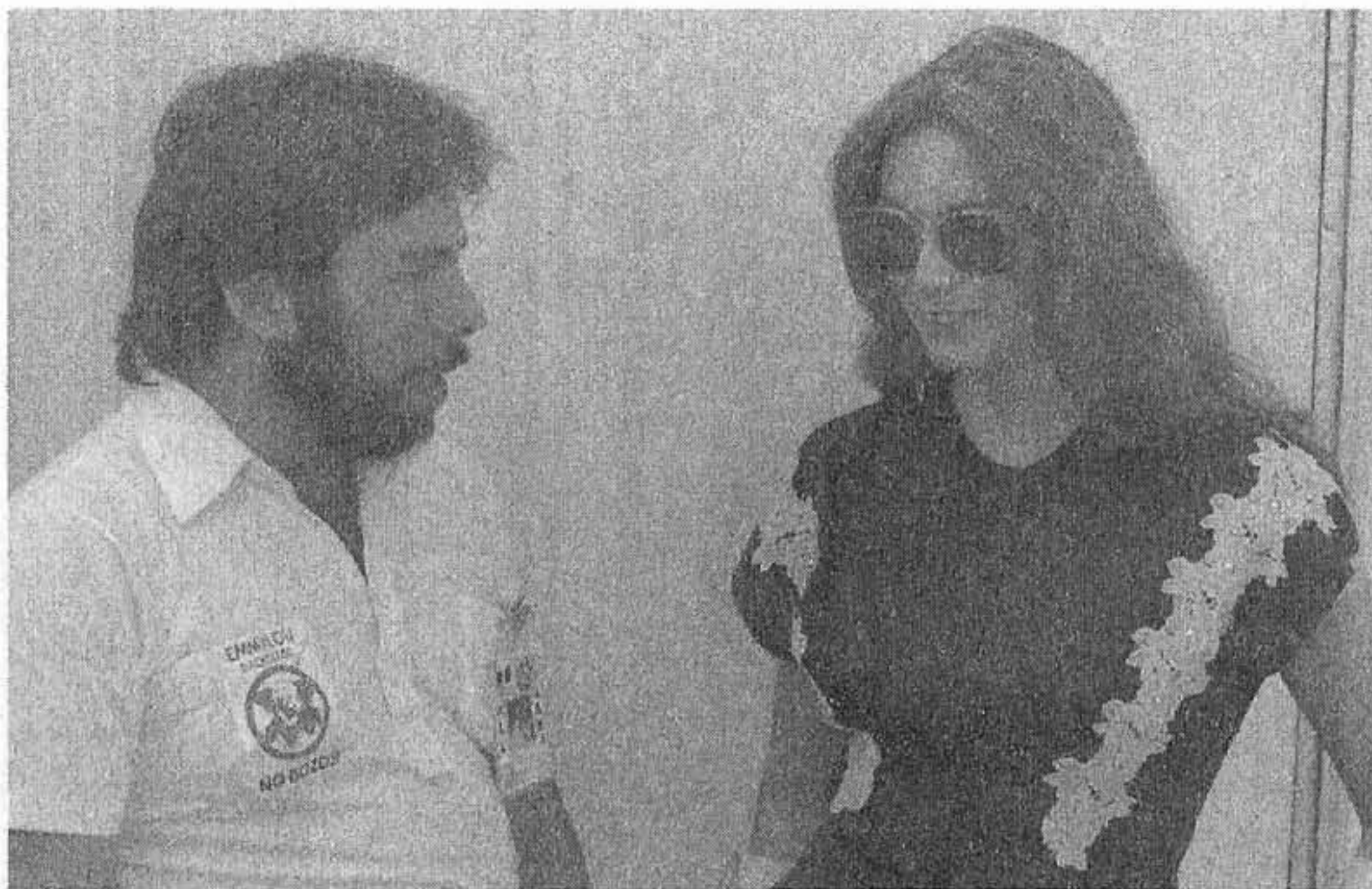
U.S. President Ronald Reagan awarded Steve and me the U.S. Technology Medal in 1985. (*Photograph courtesy of The White House*)



Here I am the day we opened up the US Festival in 1983. It lost money, but I enjoyed every minute of it. *(Photograph courtesy of Dan Sokol)*



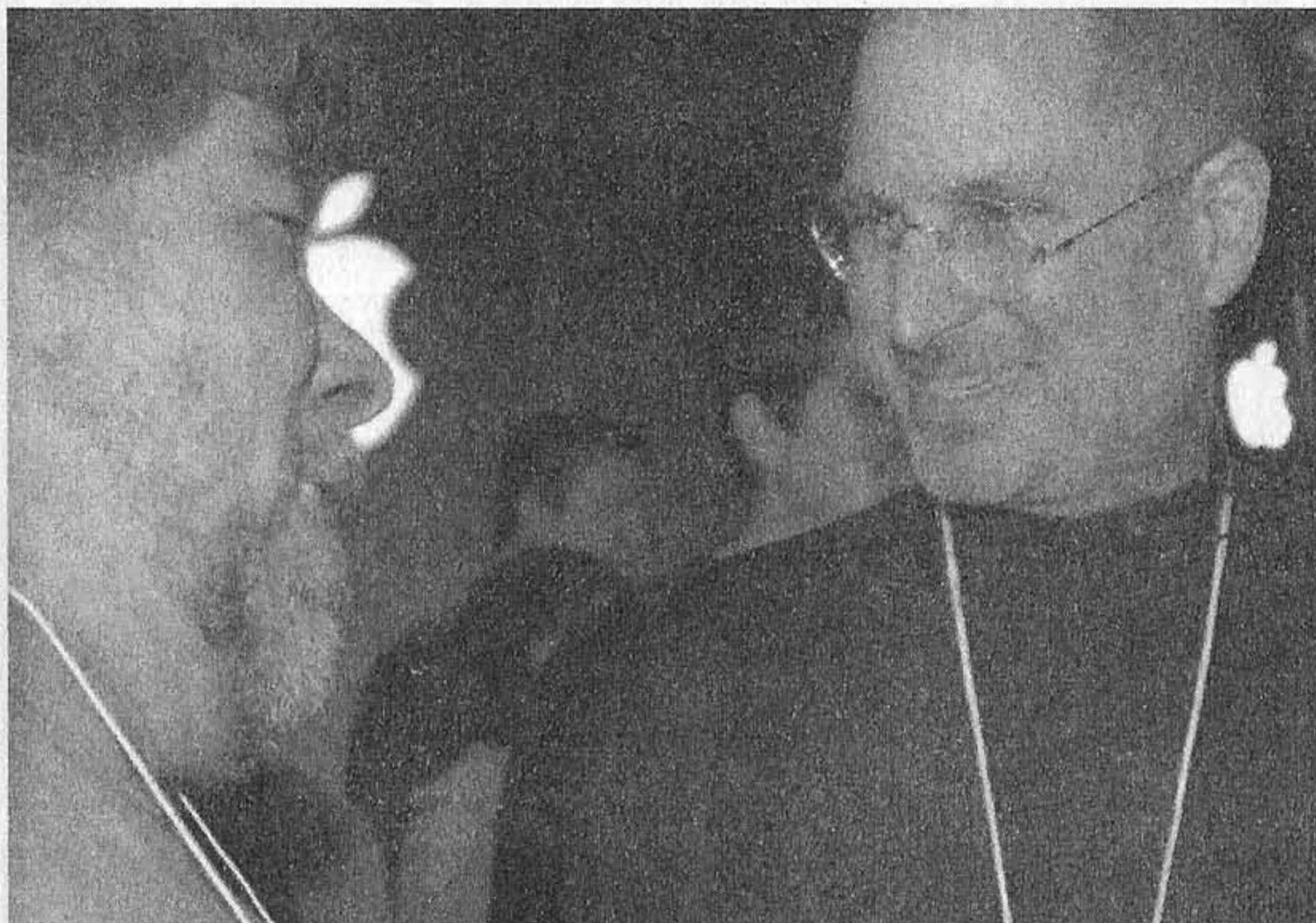
Here I am with my second wife, Candi, and then Van Halen star David Lee Roth. We were at the party before one of Van Halen's US Festival appearances. (*Photograph courtesy of Dan Sokol*)



Singer Emmylou Harris is a longtime friend of mine. She sang at my wedding and played at my US Festival. *(Photograph courtesy of Dan Sokol)*



Here I am with my second wife, Candi. She is the mother of my three children. *(Photograph courtesy of Dan Sokol)*



Steve Jobs and me sharing a laugh at Macworld 2005.

(Photograph courtesy of Alan Luckow)



Here I am with one of my Segways. I ride one all the time. It's a great personal transportation device invented by Dean Kamen. *(Photograph courtesy of Dan Sokol)*



Here I am getting ready to work with Gina in 2006.

(Photograph courtesy of Dan Sokol)

